

Mixed Reality as Human Infrastructure for Cognitive Diversity and Collective Well-Being

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Mixed Reality has advanced rapidly in terms of visual fidelity and tracking precision, but these technical milestones have not yet translated into systems that work for everyone. Current platforms often assume neurotypical behavior, treat human variability as error, and offer little transparency to clinicians, educators, or participants. I imagine a future where MR becomes a vital part of therapy, rehabilitation, and education. In this future, MR systems recognize when a child in therapy feels overwhelmed, when a patient in rehabilitation grows mentally fatigued, when a student risks losing focus during group work, and when older adults show early signs of cognitive decline. These systems adapt by drawing on the richness of human experience, creating environments that sustain participation for people with diverse needs.

Achieving this future requires MR systems that place human experience at the center. Presence, cognitive load, variability in movement, and collaboration dynamics all serve as signals that guide adaptation. Participation becomes the outcome, as systems adjust to support individuals and groups across a range of contexts. My research lays the groundwork for this trajectory. In my early studies, I manipulated realism and plausibility in MR environments and demonstrated how these factors influence presence and cognitive load [1]. I expanded this work to include distraction [2], conditioning [3], and interaction styles [4], showing consistent patterns in how humans experience shifts under different conditions. Reaction time emerged as a reliable real-time proxy for presence and load, enabling dynamic assessment beyond traditional static questionnaires. My ongoing work applies causal inference to move from patterns to mechanisms, which is critical for building systems that can sense shifts in experience and intervene at the right moments to protect both cognitive and physical well-being [5].

I also addressed the challenge of variability in human behavior. With the HoloSet dataset [6] and analyses of pose tracking fragility [7, 8], I demonstrated how natural and unpredictable movements reveal the limits of existing pipelines [9]. I developed adaptive tracking frameworks that integrate symbolic rules with neural learning, reframing variability as a resource for richer interaction [10]. This work connects directly to rehabilitation, where movements are irregular and intertwined with fatigue or recovery stages. A system that integrates motor signals with cognitive and emotional measures can provide individualized recovery programs that adapt to the patient's needs.

Collaboration adds another layer of human experience. In group settings, I began linking multimodal signals such as gaze and speech with the human perception of collaboration quality [11]. These results show how group dynamics can be sensed and interpreted [12], opening paths for environments that balance adaptation across participants. My expertise in federated learning provides the tools to design collaborative models that respect privacy [13–16], address bias [17–19], and scale across devices [20–22]. This combination will enable classrooms and therapy groups where individual support enhances rather than disrupts the shared experience.

These strands converge into a clear trajectory. My research on presence and cognitive load leads to adaptive therapy and the early detection of cognitive decline. Work on variability and tracking supports rehabilitation that integrates physical and mental resilience. Collaboration sensing and federated learning inform inclusive group environments that balance fairness and engagement. Together, these foundations define a pathway toward MR systems that make human experience their core signal and participation their ultimate outcome.

Future Directions and Expanded Impact

The first stage of my research will focus on creating proof-of-concept systems that apply the methods I have developed for sensing presence, cognitive load, variability, and collaboration to applied settings in therapy, rehabilitation, and education. These prototypes will demonstrate how MR environments can respond to signals of human experience in real time. For example, I will explore how reaction-time-based measures of presence can guide adjustments in sensory intensity for neurodivergent users. In rehabilitation, these systems will integrate motor tracking with indicators of focus and fatigue to create adaptive feedback for patient exercises. In education, they will sense an imbalance in group work and adjust pacing and turn-taking to sustain participation for all learners. In cognitive health, longitudinal models will track changes in experiential signals such as presence and variability, providing sensitive markers that can support early detection of decline. These prototypes will demonstrate how MR environments can respond to signals of human experience in real time, offering early evidence of their value in clinical and educational contexts. These exploratory studies will provide immediate evidence that MR can move beyond technical benchmarks and begin addressing the practical needs of diverse users.

Looking further ahead, my broader vision is to expand these foundations into systems that act as full partners in therapy, rehabilitation, education, and health. In the current day and age, a defining element of this work will be the integration of conversational generative AI, which can make adaptation transparent and supportive by explaining changes, scaffolding tasks, and offering reassurance. With deeper interdisciplinary training in cognitive science, neuroscience, and clinical collaboration, I will design therapy environments that adapt strategies dynamically and provide individualized pathways for children with autism, ADHD, and sensory processing differences. I will create rehabilitation systems that integrate motor, cognitive, and emotional signals into comprehensive recovery models that evolve with the patient and generalize across conditions. In classrooms, I will combine multimodal sensing with fairness-aware federated learning to balance personalization and group coherence at scale, so that students with diverse needs remain included. For aging populations, I will pair experiential signals with causal inference models to identify mechanisms of decline and design adaptive environments that promote independence.

A classroom example illustrates my vision of the future, in which students explore ecosystems in MR. Aisha, who is autistic, feels overwhelmed by complex visuals. Daniel, who has ADHD, struggles to maintain focus during group work. Traditional classrooms leave both at risk of being sidelined. In the MR environment that I envision, reaction time and gaze signals detect Aisha's overload and simplify her visuals. Daniel's drifting focus triggers adaptive pacing and conversational support from a generative AI assistant that guides him back into the task. Federated models balance contributions so both remain active participants. The teacher receives clear explanations of these adjustments, reinforcing confidence in the system. The result is a classroom where participation is supported across cognitive diversity.

This example reflects the direction of my research. By measuring presence and load, modeling variability, interpreting collaboration, and applying fairness-focused models, I have established the tools needed for MR environments that adapt to human experience. The next phase will bring these tools into therapy, rehabilitation, and education, where adaptive and inclusive systems are most urgently needed. *My long-term vision is MR as a medium that places human experience at its center and creates opportunities for participation across many cognitive and physical needs.*

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